Plate reader team

specimen cartoon although our tests will include larvae and early spat

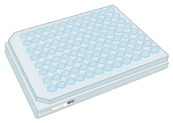
Objective:
Run resazurin assays with Pacific oysters at several temperatures

Why?
Determine if resazurin is effective in differentiating families and predicting downstream performance

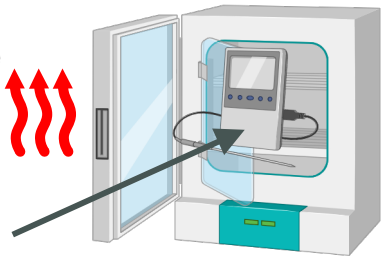
Before running trials...

- autoclave 0.2um filtered seawater
- make resazurin stock solution
- prepare required materials (review SOP)

larvae and spat trials will use mesh membrane plates

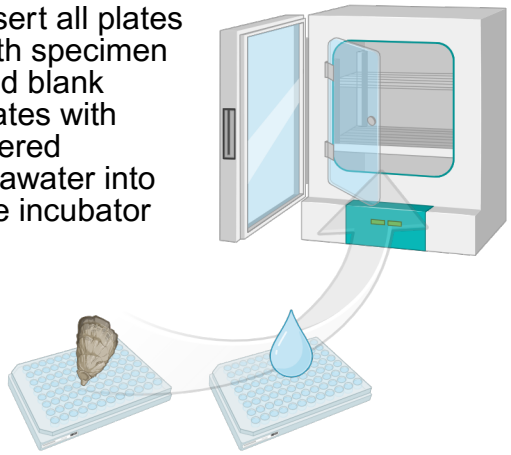


1 Program the incubator to the first temperature

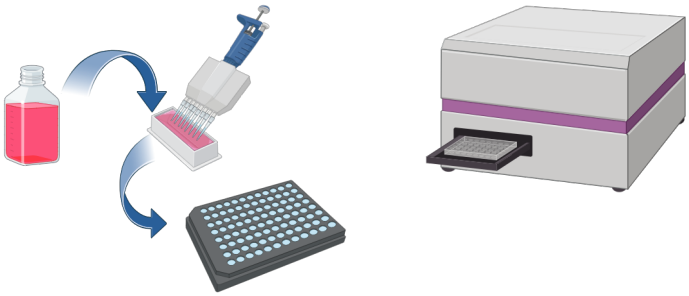


position a thermometer in the incubator to track temperature in a notebook

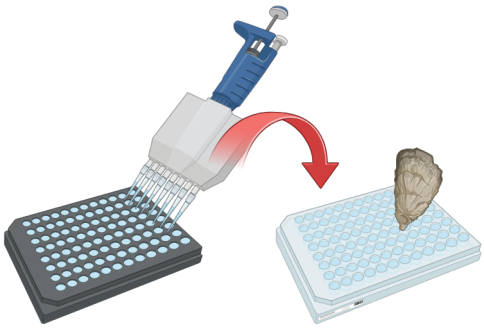
2 Insert all plates with specimen and blank plates with filtered seawater into the incubator



3 Load black plate with 300 ul resazurin, **incubate until at target temperature**, measure zeropoint flourescence

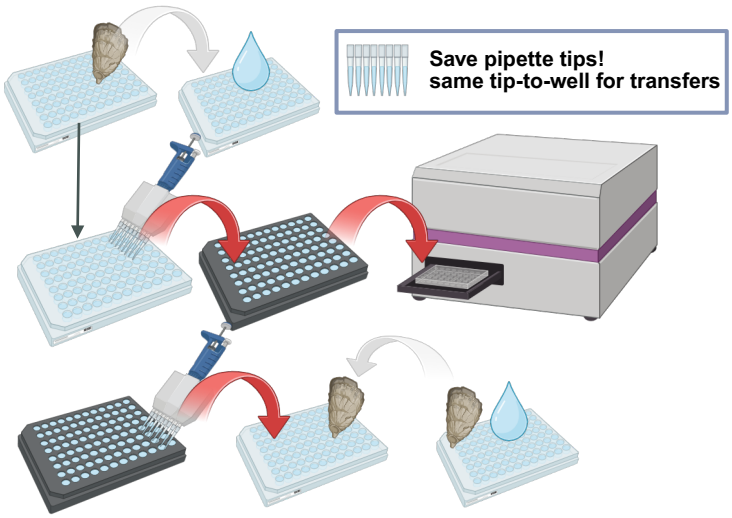


4 Pipette resazurin (same wells!) to *emptied* specimen plate #1, incubate



5 Every 20-30 minutes..

- position membrane w/specimen into seawater plate #1 and keep them incubated
- pipette resazurin from specimen plate to black bottom plate
- measure fluorescence
- pipette resazurin (**same wells!**) back into the specimen plate, position membrane w/specimen back into the specimen plate with resazurin, incubate



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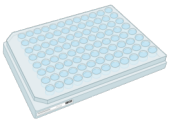
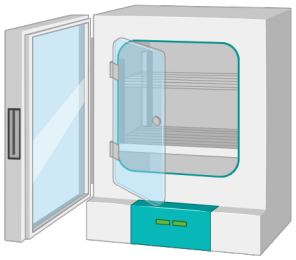
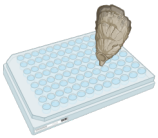


6

After **three** measurements at the target temperature

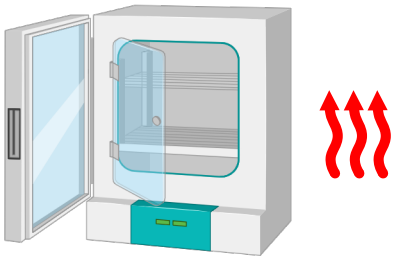
- take the membrane w/specimen on seawater plate #1, **to the microscopy team**
- dispose of the the resazurin
- throw away the black plate
- set aside the empty
- Note: do not throw away the membrane plate bottom or membrane plates, set these aside





7

Repeat steps 1, 3, 4, 5 and 6 for the remaining temperatures



Target temperatures:

- 25-26°C (hatchery temperature, starting point)
- 28°C
- 31°C
- 34°C
- 38°C

Plate loading & microscopy team

specimen cartoon although our tests will include larvae and early spat

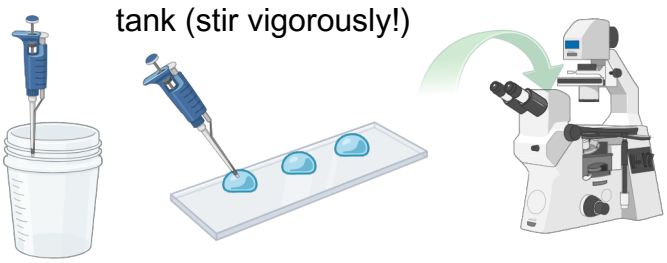
Objective:
Run resazurin assays with Pacific oysters at several temperatures

Target families based on 6 low and 6 high EBV

- #s 1-6 (high EBV)
- #s 52-58 (low EBV)

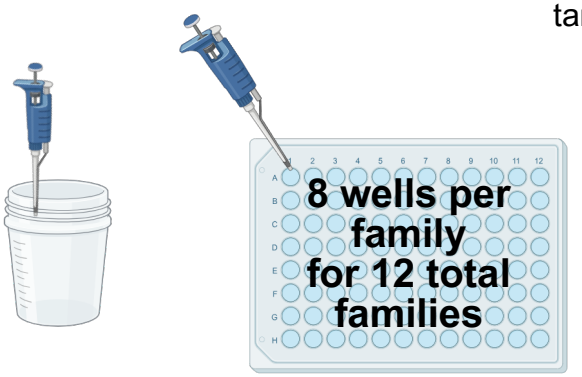
1 Hatchery team will drop tanks, screen larvae, and dilute them to 800 ml

2 Complete volumetric counts with a known volume from the 800 ml tank (stir vigorously!)



3 Calculate the required volume from the 800 ml stock to fill wells with each family

Note: for 24 hr D-hinge larvae target 100 individuals per well



4 Log the volumetric counts for each family (2) and the plate design (3) in a notebook and plate worksheets (provided)

5 Re: plates returned from after resazurin measurements
Use microscopy to identify wells..

- 'good' - near all alive and swimming normally
- 'skip/dead' - none or minimal alive
- 'maybe' - a portion alive, though some dead/deformed
  - we may dive deeper quantitatively, time permitting